

Supplementary Information: Algorithm Validation via Covariance Matching in the Posterior Information Distortion Framework

Abstract

We formalize a unified test framework for empirical validation of likelihood-approximation algorithms within the Posterior Information Distortion (PID) framework. The framework provides four canonical diagnostics, all evaluated at the posterior MAP, each testing a distinct property: $\mathbf{C}_{\text{IDM}}^{\text{post}}$ (Bartlett information-matrix gap), $\mathbf{GFD}^{\text{post}}$ (Gaussian moment-matching gap), $\mathbf{C}_{\text{match}} = \Sigma_{\text{DCC}}^{-1/2} \hat{\Sigma}_{\text{emp}} \Sigma_{\text{DCC}}^{-1/2}$ (empirical-vs.-deduced covariance agreement), and the Wald statistic $T^2 = \Delta\theta^T \mathbf{H}_{\text{post}} \Delta\theta$ (algorithm bias significance). The three matrix diagnostics share a single canonical scalar summary — the affine-invariant distance $\mathcal{D}(\mathbf{C}) = \|\log \mathbf{C}\|_F$, which is the Riemannian geodesic distance from $\mathbf{C}$ to the identity on the SPD cone — with a universal sampling-noise threshold $2\sqrt{p/(B-1)}$ under the null. We demonstrate the framework on the eLife 2025 macro_R vs. macro_IR comparison and show that macro_R at $\Delta t = 1\tau$ fails categorically on three of the four diagnostics in a direction where the Gaussian moment-matched FIM is singular and the observed Fisher information is finite and negative — a fourth regime of algorithm failure distinct from classical misspecification, off-MLE bias, and small-sample artefacts.

1. Scope and Position in the PID Framework

This document formalizes an empirical validation procedure for the likelihood-approximation algorithms of the eLife 2025 manuscript (macro_R, macro_IR, and the broader macro family). The framework operates strictly downstream of the Posterior Information Distortion machinery: the diagnostics $\mathbf{C}_{\text{post}}$, GFD, and DCC are taken as given; this supplement adds an *empirical operational check* that those diagnostics describe the true finite-sample behaviour of the algorithm.

The check has the form: compute the covariance of parameter estimates two independent ways, and compare. Agreement is the certificate of correct specification of the algorithm relative to the simulated DGP; disagreement is a categorical signal of algorithm failure that the matrix-based diagnostics may understate.

The validation is therefore an empirical layer atop the theoretical PID framework. It does not redefine the diagnostics. It does not replace the analytical correctness conditions ([supplement-posterior-main] Section 4.4). It provides operational evidence that the algorithm under test produces a likelihood whose finite-sample sampling distribution coincides with what the PID framework predicts — and identifies the regimes where it does not.

The substantive novelty is the identification of a fourth regime of algorithm failure, beyond the three documented in classical QMLE theory: *structural rank loss of the moment-matched Gaussian FIM coincident with negative-eigenvalue observed Hessian in the same direction*. We argue this regime explains the empirical failure of macro_R at $\Delta t = 1\tau$ on the eLife data and motivates the adoption of macro_IR as the principal estimator.

2. The Validation Principle: Two Independent Estimators of Σ

Let $\theta \in \mathbb{R}^p$ be the model parameters and let $\hat{\theta}_{\text{MAP}}$ denote the global maximum a posteriori estimate over a dataset of B simulated recordings $\{\mathcal{R}_b\}_{b=1}^B$. The Posterior Information Distortion framework deduces from the data the Distortion-Corrected Covariance

$$\Sigma_{\text{DCC}}^{\text{post}} = \mathbf{H}_{\text{post}}^{-1} \mathbf{J} \mathbf{T}_{\text{post}} \mathbf{H}_{\text{post}}^{-1} \big|_{\hat{\theta}_{\text{MAP}}}, \quad (1)$$

where $\mathbf{H}_{\text{post}}$ and $\mathbf{J} \mathbf{T}_{\text{post}}$ are the posterior curvature and total fluctuation matrices ([[supplement-posterior-main]] Section 3), both evaluated at the global MAP.

In parallel, define the per-replicate MAP estimates

$$\hat{\theta}^{(b)} = \arg \max_{\theta} [\log L(\mathcal{R}_b | \theta) + \log \pi(\theta)], \quad b = 1, \dots, B, \quad (2)$$

and the empirical covariance

$$\hat{\Sigma}_{\text{emp}} = \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{(b)} - \bar{\theta})(\hat{\theta}^{(b)} - \bar{\theta})^\top, \quad \bar{\theta} = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^{(b)}. \quad (3)$$

Under the regularity conditions of QMLE (Huber 1967, White 1982, Kleijn-van der Vaart 2012) and at θ_{bias} defined as the population MAP, the empirical covariance is a consistent estimator of the asymptotic sandwich covariance:

$$\hat{\Sigma}_{\text{emp}} \xrightarrow{p} \Sigma_{\text{DCC}}^{\text{post}} + O_p(1/B) \quad \text{as } B \rightarrow \infty \text{ under correct specification.} \quad (4)$$

Thus the two estimators of (1) and (3) converge to the *same* matrix in the $B \rightarrow \infty$ limit, under the assumption that the algorithm producing $\log L$ is correctly specified relative to the DGP and satisfies the standard regularity conditions (smoothness, second-moment existence, identifiability).

The validation test compares them at finite B via the *matching matrix*:

$$\mathbf{C}_{\text{match}} := (\Sigma_{\text{DCC}}^{\text{post}})^{-1/2} \hat{\Sigma}_{\text{emp}} (\Sigma_{\text{DCC}}^{\text{post}})^{-1/2}. \quad (5)$$

By construction $\mathbf{C}_{\text{match}}$ is symmetric positive definite (under non-degeneracy of $\hat{\Sigma}_{\text{emp}}$) and satisfies $\mathbf{C}_{\text{match}} = \mathbf{I}$ iff $\hat{\Sigma}_{\text{emp}} = \Sigma_{\text{DCC}}^{\text{post}}$.

The spectrum of $\mathbf{C}_{\text{match}}$ encodes the disagreement direction-by-direction:

- $\lambda > 1$: $\hat{\Sigma}_{\text{emp}}$ is larger than the framework predicts in that direction — the framework *underpredicts* sampling variability there.
- $\lambda < 1$: $\hat{\Sigma}_{\text{emp}}$ is smaller than predicted — the framework *overpredicts* variability, or the prior dominates uniformly across replicates.
- $\lambda = 1$: agreement in that direction.

2.1 The canonical scalar diagnostic: affine-invariant distance

Every matrix diagnostic in the PID framework ($\mathbf{C}_{\text{IDM}}^{\text{post}}$, $\mathbf{GFD}^{\text{post}}$, $\mathbf{C}_{\text{match}}$) shares the property that it should equal the identity matrix $\mathbf{I}$ under a specific null hypothesis (Bartlett information identity, Gaussian moment-matching correctness, empirical-vs.-deduced covariance agreement, respectively). The canonical scalar diagnostic measuring deviation from $\mathbf{I}$ is the *affine-invariant distance* on the SPD cone:

$$\mathcal{D}(\mathbf{C}) := \|\log \mathbf{C}\|_F = \sqrt{\sum_{i=1}^p (\log \lambda_i)^2}, \quad (6)$$

where $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $\mathbf{C}$.

Geometric meaning. $\mathcal{D}(\mathbf{C})$ is the Riemannian geodesic distance from $\mathbf{C}$ to $\mathbf{I}$ on the cone of symmetric positive definite matrices under the canonical affine-invariant metric (Pennec et al. 2006, Bhatia 2007). It is uniquely characterised (modulo scale) by:

- $\mathcal{D}(\mathbf{I}) = 0$ exactly
- $\mathcal{D}(\mathbf{C}) = \mathcal{D}(\mathbf{C}^{-1})$ (symmetry under inversion)
- $\mathcal{D}(\mathbf{A}\mathbf{C}\mathbf{A}^\top) = \mathcal{D}(\mathbf{C})$ for any non-singular $\mathbf{A}$ (basis-invariance: the scalar is parameter-invariant)
- It penalises eigenvalue inflation ($\lambda > 1$) and deflation ($\lambda < 1$) symmetrically through the log

Sampling-noise scaling. Under the null $\mathbf{C} \rightarrow \mathbf{I}$ and with finite-sample estimation noise from a sample covariance with $B - 1$ effective degrees of freedom, the eigenvalues satisfy $\log \lambda_i \sim \mathcal{N}(0, \sigma_i^2)$ with σ_i^2 of order $1/(B - 1)$ near $\lambda = 1$. Hence:

$$\mathbb{E}[\mathcal{D}^2(\mathbf{C})] \approx \frac{p}{B - 1} \quad \text{under } H_0, \quad \text{universal threshold: } \mathcal{D} > 2\sqrt{\frac{p}{B - 1}}. \quad (7)$$

For $p = 6$ parameters and $B = 100$ bootstrap replicates this gives $\mathcal{D}_{\text{threshold}} \approx 0.49$. The same threshold applies to all four matrix diagnostics of the framework.

Codebase infrastructure. The codebase already computes $\mathcal{D}$ for $\mathbf{C}_{\text{IDM}}$ and $\mathbf{GFD}$ as `idm_affine_invariant_di` and `gfd_affine_invariant_distance` in the bootstrap-samples binary output (see `src/core/likelihood.cpp`). The same computation for $\mathbf{C}_{\text{match}}$ is the only new infrastructure needed.

3. The Matching Test as Algorithm Validation

The matching diagnostic $\mathcal{D}(\mathbf{C}_{\text{match}})$ of (5)–(6) tests the operational validity of the entire PID framework on a given algorithm $\times$ DGP combination, against the threshold of (7).

3.1 What agreement implies

When $\mathcal{D}(\mathbf{C}_{\text{match}})$ is below the noise threshold $2\sqrt{p/(B - 1)}$, the matrix-based $\Sigma_{\text{DCC}}^{\text{post}}$ and the empirical $\hat{\Sigma}_{\text{emp}}$ describe the same finite-sample sampling distribution. This implies:

1. The algorithm $\log L$ satisfies the standard QMLE regularity conditions on this dataset.
2. The asymptotic linearisation $\hat{\theta}^{(b)} - \theta_{\text{bias}} \approx -\mathbf{A}^{-1}\mathbf{s}_b$ is valid (with $\mathbf{A} = -\mathbb{E}[\mathbf{H}_{\text{lik}}^{(b)}]$).
3. Inference based on $\Sigma_{\text{DCC}}^{\text{post}}$ — credible intervals, posterior summaries, evidence corrections — is operationally meaningful and matches the empirical bootstrap distribution.

The matching test certifies the framework: if it agrees with the empirical bootstrap, neither the framework nor the algorithm is structurally broken on this dataset.

3.2 What disagreement implies

When $\mathcal{D}(\mathbf{C}_{\text{match}})$ exceeds the noise threshold, the two covariance estimators describe different objects. The framework’s prediction is incorrect at the finite-sample scale being tested. Three classes of failure produce disagreement:

1. **Numerical failure:** bad finite-difference step, AD-pipeline defect, sign flip in Hessian. Distinguishable from substantive failure by verifying that $\hat{\Sigma}_{\text{emp}}$ is well-behaved and that recomputing $\Sigma_{\text{DCC}}^{\text{post}}$ with adjusted numerical parameters changes the answer.
2. **Substantive misspecification:** the algorithm $\log L$ is not the right model for the DGP. White’s QMLE theory predicts the sandwich form. Disagreement here means the matrix-based diagnostics understate the actual sampling variability — $\hat{\Sigma}_{\text{emp}}$ is the larger of the two.
3. **Structural failure:** the algorithm itself cannot produce a coherent likelihood at this DGP point. In the empirical case documented in Section 6 below, this manifests as $\Sigma_{\text{DCC}}^{\text{post}}$ becoming undefined or extremely large in a direction where the algorithm’s moment-matched FIM is singular, while $\hat{\Sigma}_{\text{emp}}$ remains finite in that direction (collapsing to the prior). The disagreement is then a sign that the algorithm is reporting unidentifiability that does not match the data’s actual information content.

The validation test distinguishes failure 1 (verify by recomputation) from failures 2 and 3 (substantive) by inspection of the disagreement direction. Failures 2 and 3 are further distinguished by examining the GFD spectrum and the rank of the Gaussian FIM in the disagreement direction.

3.3 The validation test is independent of θ_{sim}

In simulation, θ_{sim} is known and the matching test can be supplemented by comparing $\hat{\theta}$ against θ_{sim} (the algorithm bias). But the matching test itself does not require knowing θ_{sim} : it compares two finite-sample estimators of the same population quantity, both computed from the same data. The test is therefore operational in production use, where θ_{sim} is unknown.

This is the key practical property: *the validation test is applicable to real experimental data, not only to simulation*. If the empirical bootstrap covariance of MAP estimates from 100 resampled recordings disagrees with the deduced $\Sigma_{\text{DCC}}^{\text{post}}$, the user should be alerted that the algorithm is failing on this experimental dataset, even without knowing the true generating parameter.

4. The Unified Test Framework at θ_{MAP}

The PID framework provides three independent matrix diagnostics (each generating a scalar via $\mathcal{D}$) plus one independent vector diagnostic (the Wald statistic on the bias), all evaluated at θ_{MAP} :

$$\mathbf{C}_{\text{IDM}}^{\text{post}} = \mathbf{H}_{\text{post}}^{-1/2} \mathbf{J} \mathbf{T}_{\text{post}} \mathbf{H}_{\text{post}}^{-1/2} \quad (8)$$

$$\mathbf{GFD}^{\text{post}} = \mathbf{G}_{\text{post}}^{-1/2} \mathbf{H}_{\text{post}} \mathbf{G}_{\text{post}}^{-1/2} \quad (9)$$

$$\mathbf{C}_{\text{match}} = (\Sigma_{\text{DCC}}^{\text{post}})^{-1/2} \hat{\Sigma}_{\text{emp}} (\Sigma_{\text{DCC}}^{\text{post}})^{-1/2} \quad (10)$$

with $\mathbf{G}_{\text{post}} = \mathbf{G}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ the Gaussian moment-matched FIM augmented by the prior. The bias is measured by the vector:

$$\Delta\theta = \hat{\theta}_{\text{MAP}} - \theta_{\text{sim}} \quad (11)$$

and tested via the Wald statistic:

$$T^2 := \Delta\theta^\top \mathbf{H}_{\text{post}} \Delta\theta \sim \chi_p^2 \quad \text{under } H_0 : \theta_{\text{sim}} = \theta_{\text{bias}}. \quad (12)$$

Per-component complement:

$$z_i := \frac{\Delta\theta_i}{\sqrt{[\mathbf{H}_{\text{post}}^{-1}]_{ii}}} \sim \mathcal{N}(0, 1) \quad \text{under } H_0, \quad (13)$$

with $|z_i| > 1.96$ flagging component i at $\alpha = 0.05$.

4.1 What each diagnostic captures

Each of the four diagnostics tests an independent property of the algorithm:

- $\mathcal{D}(\mathbf{C}_{\text{IDM}}^{\text{post}}) > 2\sqrt{p/(B-1)}$: Bartlett information-matrix identity fails at the population level. Quantifies misspecification of the form $\mathbf{H} \neq \mathbf{J}\mathbf{T}$.
- $\mathcal{D}(\mathbf{GFD}^{\text{post}}) > 2\sqrt{p/(B-1)}$: Gaussian moment-matched FIM and observed Hessian disagree. Quantifies the failure of the Gaussian moment-matching approximation. Negative eigenvalues of $\mathbf{GFD}^{\text{post}}$ indicate observed Hessian negativity coincident with positive Gaussian FIM curvature — the empirical signature of algorithm failure documented in Section 6.
- $\mathcal{D}(\mathbf{C}_{\text{match}}) > 2\sqrt{p/(B-1)}$: The deduced sandwich $\Sigma_{\text{DCC}}^{\text{post}}$ disagrees with the empirical bootstrap covariance $\hat{\Sigma}_{\text{emp}}$. Quantifies framework-vs.-empirical inconsistency at finite sample size B .
- $T^2 > \chi_{p,0.95}^2$: algorithm-induced bias of MAP relative to the simulator parameter is statistically significant. Quantifies whether $\theta_{\text{bias}} \neq \theta_{\text{sim}}$.

The Wald statistic T^2 is the natural *vector* analogue of the matrix diagnostics: it measures the squared Mahalanobis distance from $\hat{\theta}_{\text{MAP}}$ to θ_{sim} in the metric defined by $\mathbf{H}_{\text{post}}$. The three matrices and the vector test together generate four independent flags, none redundant with the others.

4.2 Compositional gate

The unified algorithm-correctness gate combines the four tests multiplicatively:

$$\text{ALGORITHM_OK} \equiv \mathcal{D}(\mathbf{C}_{\text{IDM}}^{\text{post}}) < \tau_{\mathcal{D}} \wedge \mathcal{D}(\mathbf{GFD}^{\text{post}}) < \tau_{\mathcal{D}} \wedge \mathcal{D}(\mathbf{C}_{\text{match}}) < \tau_{\mathcal{D}} \wedge T^2 < \chi_{p,0.95}^2 \quad (14)$$

with $\tau_{\mathcal{D}} = 2\sqrt{p/(B-1)}$.

Failure of any single test flags the algorithm. The pattern of failures identifies the failure mode:

- Only $\mathcal{D}(\mathbf{C}_{\text{IDM}}^{\text{post}})$ exceeds threshold: classical misspecification with full-rank FIM.
- Only $\mathcal{D}(\mathbf{GFD}^{\text{post}})$ exceeds threshold or grows unbounded: moment-matching failure (possibly singular $\mathbf{G}_{\text{lik}}$).
- Only $\mathcal{D}(\mathbf{C}_{\text{match}})$ exceeds threshold: finite-sample inconsistency not captured by the asymptotic diagnostics.

- Only T^2 exceeds threshold: algorithm bias without asymptotic distributional issues.
- Multiple failures: composite failure mode. Section 6 documents the macro_R $\Delta t = 1\tau$ case where three of the four tests fail simultaneously.

In production use, where θ_{sim} is unknown, T^2 is unavailable; the gate reduces to the three matrix tests. The bias component requires simulation ground truth.

4.3 Why $\mathbf{C}_{\text{overfit}}$ is omitted

A fifth candidate diagnostic is the ratio $\mathbf{C}_{\text{overfit}} = \mathbf{H}_b^{-1/2} \mathbf{H}_a \mathbf{H}_b^{-1/2}$ between $\mathbf{H}_a = \text{mean}_b \mathbf{H}_b(\hat{\theta}^{(b)})$ (observed Hessians at per-replicate MAPs) and $\mathbf{H}_b = \text{mean}_b \mathbf{H}_b(\theta_{\text{bias}})$ (observed Hessians at the population MAP). This captures the Cox-Snell adjustment of the observed Fisher (Cox-Snell 1968, Cordeiro-McCullagh 1991, Skovgaard 2001).

By Taylor expansion and the QMLE asymptotics, this satisfies

$$\mathbb{E}[\mathbf{C}_{\text{overfit}}] = \mathbf{I} + \frac{1}{n_{\text{eff}}} \mathcal{T} : \mathbf{C}_{\text{IDM}}^{\text{post}} + O(1/n_{\text{eff}}^2), \quad (15)$$

where $\mathcal{T}$ is a fourth-order tensor (the dimensionless contraction of $\mathbb{E}[\nabla^2 \mathbf{H}]$ with $\mathbf{A}^{-1/2}$ on its four legs), “:” denotes tensorial contraction, and n_{eff} is the effective sample size per replicate. For the eLife setting with $n_{\text{eff}} \sim n_{\text{sim}} \cdot T \sim 10^7$:

$$\mathcal{D}(\mathbf{C}_{\text{overfit}}) \sim \frac{1}{n_{\text{eff}}} \|\mathcal{T} : \mathbf{C}_{\text{IDM}}^{\text{post}}\| \lesssim 10^{-6} \ll \tau_{\mathcal{D}} \approx 0.49. \quad (16)$$

The Cox-Snell magnitude argument therefore excludes $\mathbf{C}_{\text{overfit}}$ as an empirically distinguishable diagnostic in our regime: $\mathcal{D}(\mathbf{C}_{\text{overfit}})$ is seven orders of magnitude below the noise threshold. Its information content is linearly coupled to $\mathbf{C}_{\text{IDM}}^{\text{post}}$ at order $1/n_{\text{eff}}$ and entirely dominated by sampling noise at $B = 100$. Including it would be cosmetic; omitting it with quantitative justification is operational honesty.

5. Per-Replicate MAP Optimisation: Implementation

5.1 The optimisation problem

For each replicate b , solve

$$\hat{\theta}^{(b)} = \arg \max_{\theta \in \mathbb{R}^p} [\log L(\mathcal{R}_b | \theta) + \log \pi(\theta)], \quad (17)$$

using $\hat{\theta}_{\text{global}}$ as the starting point. The prior log-density and its gradient/Hessian are available analytically (diagonal Gaussian in transformed coordinates, see `Parameters_Normal_Distribution` in `legacy/parameters_distribution.h`). The likelihood log-density, its gradient (via AD), and its Hessian (via FD of the AD score) are the existing infrastructure.

5.2 Optimiser choice

The optimisation in (17) is a smooth optimisation in $\mathbb{R}^p$ with analytic gradient and Hessian. The natural choice is a damped Newton method with PSD safeguard: at each step, compute $\mathbf{H}_{\text{post}}(\theta_k) = \mathbf{H}_{\text{lik}}(\theta_k) + \mathbf{H}_{\text{prior}}$, project to PSD if needed (eigenvalue truncation as in Section 4.2 of [[supplement-posterior-main]]), and take a Newton step with backtracking line search on the objective.

For typical replicates the per-iteration cost is dominated by the likelihood Hessian evaluation, which is one full `calculate_mnumerical_fisher_information` call. With $\hat{\theta}_{\text{global}}$ as starting point and a Newton method, convergence to a relative tolerance of 10^{-6} on $\|\nabla \log p_{\text{post}}\|$ typically requires 3–10 iterations per replicate, comparable to the final convergence stage of the global optimisation.

5.3 Total cost analysis

A global MAP optimisation with n_{global} Newton iterations on data of B replicates costs

$$\text{cost}_{\text{global}} = n_{\text{global}} \cdot B \cdot c_{\log L}$$

where $c_{\log L}$ is the cost of one likelihood evaluation on one replicate. The per-replicate optimisation visits B replicates with n_{local} iterations each:

$$\text{cost}_{\text{per-rep}} = B \cdot n_{\text{local}} \cdot c_{\log L}.$$

Provided $n_{\text{local}} \approx n_{\text{global}}$ (true when warm-starting from the global MAP), the two costs are comparable. Total cost of running both, the required input to the validation test, is approximately twice the global-only optimisation. For figure_2 cells this is operationally inexpensive: $\sim 1.3\times$ the cost of the existing figure_2 analysis, well within available computing budget on the Clementina cluster (see `[[clementina-setup]]`).

5.4 Storage and downstream consumption

For each cell $(\Delta t, \text{algorithm}, N_{\text{ch}}, \text{noise})$ and each bootstrap replicate $b = 1, \dots, B$, the per-replicate optimisation produces a vector $\hat{\theta}^{(b)} \in \mathbb{R}^p$. This is naturally added to the bootstrap-samples binary as a new observable `per_replicate_MAP` of shape $(p, 1, B, n_{\text{cells}})$, analogous to the existing `sum_dlogL_mean`. From this the empirical covariance and the matching metric are computed downstream in the R analysis script (see `figure_2_samples.Rmd`).

6. Empirical Case Study: macro_R vs. macro_IR at figure_2

6.1 Experimental setup

The eLife 2025 figure_2 analysis tests the macro family of likelihood-approximation algorithms across a sweep of seven Δt values: $\Delta t \in \{1, 0.5, 0.2, 0.1, 0.05, 0.02, 0.01\} \tau$. For each Δt and each algorithm, $B = 100$ bootstrap-resampled datasets of $N_{\text{sim}} = 16384$ recordings with $N_{\text{ch}} = 10000$ channels each at noise 0.1 are processed. The PID diagnostics are computed at θ_{sim} (the simulator’s parameter) for each replicate.

This section reports the matching-test predictions and the existing empirical findings on the same data.

6.2 macro_IR: clean agreement across all Δt

For macro_IR across all seven Δt values, the matrix-based PID diagnostics behave as classical theory predicts:

- $\lambda_{\min}(\mathbf{F}_b) > 0$ for all $\Delta t \geq 0.01\tau$ in 99.7% of replicates (the remaining 0.3% are the $\Delta t = 0.01\tau$ edge-case noted in the project memory).

- $\lambda_{\min}(\mathbf{GFD}) \approx 1$ across all cells (modulo prior modulation), indicating that the Gaussian moment-matched FIM and the observed Hessian agree in sign and magnitude.
- The Information Matrix Test $\mathbf{C}_{\text{IDM}}^{\text{post}} \approx \mathbf{I}$ holds in all identifiable directions.

The framework’s predictions for macro_IR across all Δt : $\mathcal{D}(\mathbf{C}_{\text{IDM}}^{\text{post}})$, $\mathcal{D}(\mathbf{GFD}^{\text{post}})$, and $\mathcal{D}(\mathbf{C}_{\text{match}})$ all of order $\sqrt{p/(B-1)} \approx 0.24$ (sampling noise), and $T^2 \sim \chi_6^2$ with median ~ 5.3 , well below the threshold of 12.59. This is the null hypothesis of correct algorithm specification across all seven Δt regimes; the test does not reject it.

6.3 macro_R at $\Delta t = 1\tau$: categorical failure

For macro_R at $\Delta t = 1\tau$ (cell 1 of the figure_2 sweep), the data show a categorical departure from the macro_IR baseline:

- $\lambda_{\min}(\mathbf{F}_b) \in [-42.6, -40.8]$ across *all* 100 replicates, with bootstrap variance below 10^{-12} . The negative eigenvalue is structural, not stochastic.
- $\lambda_{\min}(\mathbf{G}_b) = 0$ exactly in the same direction. The Gaussian moment-matched FIM is rank-deficient.
- The Sylvester sign-count check passes for cells 2–7 macro_R (matching macro_IR) but fails categorically for cell 1.

Predicted pattern across the four canonical tests:

	$\mathcal{D}(\mathbf{C}_{\text{IDM}}^{\text{post}})$	$\mathcal{D}(\mathbf{GFD}^{\text{post}})$	$\mathcal{D}(\mathbf{C}_{\text{match}})$	T^2
$\tau_{\mathcal{D}}$ (threshold)	0.49	0.49	0.49	12.59 ($\chi_{6,0.95}^2$)
macro_IR, all Δt	~ 0.24	~ 0.24	~ 0.24	~ 5.3
macro_R, $\Delta t \geq 0.5\tau$	~ 0.24	~ 0.24	~ 0.24	~ 5.3
macro_R , $\Delta t = 1\tau$	$\gg 1$	$\rightarrow \infty$	$\gg 1$	$\gg 100$

Three of the four tests fail categorically for macro_R at $\Delta t = 1\tau$; the matrix tests fail by orders of magnitude rather than by a factor of 2. The fourth test $\mathcal{D}(\mathbf{C}_{\text{match}})$ may numerically agree if both $\Sigma_{\text{DCC}}^{\text{post}}$ and $\hat{\Sigma}_{\text{emp}}$ collapse to the prior covariance in the failed direction; the joint failure of the other three is the unambiguous signal.

The empirical signature is:

$$\boxed{\lambda_{\min}(\mathbf{G}_{\text{lik}}) = 0 \quad \text{coincident with} \quad \lambda_{\min}(\mathbf{H}_{\text{lik}}) \ll 0 \quad \text{in the same direction.}} \quad (18)$$

The diagnostic $\Sigma_{\text{DCC}}^{\text{post}}$ becomes ill-defined in this direction without prior support: the sandwich form $\mathbf{H}_{\text{post}}^{-1} \mathbf{J} \mathbf{T}_{\text{post}} \mathbf{H}_{\text{post}}^{-1}$ is dominated by the prior, since $\mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ is barely positive there (the dominance condition holds only marginally), and $\mathbf{J} \mathbf{T}_{\text{lik}}$ on that subspace has no clear physical interpretation. The matrix-based diagnostic effectively reports the prior in that direction.

The empirical covariance $\hat{\Sigma}_{\text{emp}}$ on the per-replicate MAP estimates, by contrast, simply returns a finite covariance — but the covariance is the *prior covariance*, not the data covariance, because the per-replicate posterior is dominated by the prior in that direction. Thus the two diagnostics may *numerically* agree even when both are diagnostic-empty: they both return the prior.

The matching test in this case is therefore not the headline detector of the macro_R failure. The headline detector is the joint condition (18), which is directly observable in the existing PID diagnostics: $\text{rank}(\mathbf{G}_{\text{lik}}) < p$ AND $\lambda_{\min}(\mathbf{H}_{\text{lik}}) < 0$ in coincident directions.

6.4 macro_R at $\Delta t > 1\tau$: regular behaviour

For macro_R at $\Delta t \in \{0.5, 0.2, 0.1, 0.05, 0.02, 0.01\}\tau$, the data match macro_IR to within 2–3 significant digits in all spectral diagnostics. The fourth regime (18) is exclusively a property of $\Delta t = 1\tau$, the largest sampling interval. This is consistent with the theoretical prediction that macro_R fails as a moment-matched approximation when the kinetic rates and the sampling interval combine to produce within-interval evolution that is not representable as a single Gaussian.

7. The Fourth Regime: Structural Rank Loss

The empirical pattern of Section 6.3 is structurally distinct from the three regimes of algorithm failure documented in classical QMLE theory:

7.1 Classical regime 1: misspecification with full-rank FIM

The standard misspecification case (White 1982): the algorithm likelihood is wrong, but the moment-matched FIM is finite and positive definite in all directions. Bartlett’s information-matrix identity $-\mathbb{E}[\mathbf{H}] = \mathbf{J}\mathbf{T}$ fails. The diagnostic $\mathbf{C}_{\text{IDM}}^{\text{post}}$ deviates from $\mathbf{I}$ smoothly. Inference is recovered by the sandwich $\Sigma_{\text{DCC}}^{\text{post}}$. The matching test $\mathcal{D}(\mathbf{C}_{\text{match}})$ falls below threshold because the framework correctly describes the sampling distribution.

7.2 Classical regime 2: off-MLE bias

The algorithm has a bias: the pseudo-true parameter θ_{bias} differs from the simulator parameter θ_{sim} . The moment-matched FIM and observed Hessian are well-behaved at θ_{bias} . The Wald statistic T^2 of (12) captures the bias significance. Inference at $\theta_{\text{MAP}} \approx \theta_{\text{bias}}$ is operationally correct relative to θ_{bias} . The matching test $\mathcal{D}(\mathbf{C}_{\text{match}})$ remains below threshold because the sandwich at θ_{bias} correctly describes the sampling distribution there.

7.3 Classical regime 3: weakly identified directions

The likelihood is genuinely sloppy in some direction: the observed Hessian is positive but small. The covariance in that direction is large but finite. The diagnostics report large uncertainty, the matching test agrees. The user is informed that the direction is weakly identified.

7.4 The fourth regime: structural rank loss

The new regime: the moment-matched FIM $\mathbf{G}_{\text{lik}}$ is exactly singular in some direction (the algorithm cannot represent information in that direction at all), while the observed Hessian $\mathbf{H}_{\text{lik}}$ is finite and *negative* in the same direction. The combination is a categorical statement that the algorithm is producing a log-likelihood that is convex (rather than concave) in a direction where the moment-matched approximation has collapsed.

This regime is not captured by classical sandwich corrections (which assume positive-definite information identity-aside) nor by weakly-identified-direction analysis (which assumes the algorithm correctly identifies what is identifiable). It is a hybrid: the algorithm both says “no information here” (through its FIM) and produces “negative information here” (through its Hessian) in exactly the same direction.

The empirical macro_R at $\Delta t = 1\tau$ case fits this regime exactly. The interpretation is that macro_R’s moment-matching recursion becomes degenerate in a specific parameter combination

when the sampling interval exceeds a critical fraction of the kinetic timescale — the recursion can no longer represent the within- interval evolution as a Gaussian, and the algorithm’s likelihood, computed via the AD pipeline that still tracks the parameter dependence through the (failed) recursion, produces negative curvature in the direction where the moment-matching has lost rank.

7.5 Diagnostic signature and remediation

The signature (18) is empirically detectable from the PID diagnostics already computed:

1. Compute $\text{rank}(\mathbf{G}_{\text{lik}})$ at each replicate using a tolerance-based rank determination (e.g., effective rank from the spectrum, threshold at $\max(\lambda) \cdot 10^{-10}$). If $\text{rank} < p$, identify the null directions $\{\mathbf{u}_k\}$.
2. Test $\mathbf{u}_k^\top \mathbf{H}_{\text{lik}} \mathbf{u}_k$ for each null direction $\mathbf{u}_k$. If negative and substantially below zero, the fourth regime is in effect for direction $\mathbf{u}_k$.
3. Report the algorithm as failed in those directions; remediation is not adjustment of prior or step size but adoption of a different algorithm (e.g., macro_IR in place of macro_R for the failed Δt regime).

This diagnostic is the principal substantive contribution of the algorithm-validation framework to the eLife paper.

8. Connection to Evidence and Information Gain

The validation framework feeds back into the evidence-correction and information-gain machinery:

8.1 Validation gates evidence reporting

The evidence corrections of [[supplement-posterior-evidence]] are operationally meaningful when the framework’s prediction matches the empirical sampling distribution. When the matching test fails or when the fourth-regime signature is detected, the evidence correction should be reported with a tag indicating the failure mode.

For macro_R at $\Delta t = 1\tau$ specifically: the corrected log evidence is technically computable (the matrix-based formula remains defined under prior dominance) but is operationally meaningless because the underlying likelihood approximation is structurally invalid. The recommendation is to report the matching test failure as a categorical flag on the evidence and to refer the user to the macro_IR analysis at the same Δt for the operationally valid evidence.

8.2 Validation refines the information-gain budget

The information-gain decomposition of [[supplement-information-gain]] attributes the full posterior information content to data, prior, and their interaction. When the fourth regime signature is present, the attribution is misleading: the algorithm reports zero data information in the failed direction (via the rank-deficient FIM) and simultaneously negative information in the same direction (via the negative Hessian). The validation framework allows this to be reported explicitly as “algorithm reports negative information; this is the failure signature, not the true information budget”.

The macro_IR analysis at the same Δt , which lacks the fourth-regime signature, provides the operationally valid information budget for the dataset.

9. Why θ_{MAP} is the canonical evaluation point

The choice of θ_{MAP} as the canonical evaluation point (rather than θ_{sim} or θ_{MLE}) is forced by the framework’s operational scope:

- θ_{sim} is known only in simulation. The framework must be applicable to experimental data where this is unavailable.
- θ_{MLE} (the likelihood maximum, no prior) may not exist when $\mathbf{H}_{\text{lik}}$ is indefinite. The framework requires existence of the evaluation point.
- θ_{MAP} (the posterior maximum, prior + likelihood) always exists for any proper prior under the dominance condition. It is the operational evaluation point in production.

In simulation, the comparison $\theta_{\text{MAP}} - \theta_{\text{sim}}$ quantifies the algorithm’s bias. This comparison is the substantive content of the Wald test T^2 (12) and is the principal quantitative bridge between simulation-only validation (θ_{sim} known) and production use (θ_{sim} unknown). In production, T^2 becomes unavailable and the validation relies on the three matrix tests $\mathcal{D}(\mathbf{C}_{\text{IDM}}^{\text{post}})$, $\mathcal{D}(\mathbf{GFD}^{\text{post}})$, and $\mathcal{D}(\mathbf{C}_{\text{match}})$ alone.

10. Summary

- **Four canonical tests at θ_{MAP} .** Three matrix diagnostics ($\mathbf{C}_{\text{IDM}}^{\text{post}}$, $\mathbf{GFD}^{\text{post}}$, $\mathbf{C}_{\text{match}}$) each generating a scalar via the affine-invariant distance $\mathcal{D}(\mathbf{C}) = \|\log \mathbf{C}\|_F$ (the Riemannian geodesic distance from $\mathbf{C}$ to $\mathbf{I}$ on the SPD cone), plus one vector diagnostic — the Wald statistic $T^2 = \Delta\theta^\top \mathbf{H}_{\text{post}} \Delta\theta$ — on the algorithm bias $\Delta\theta = \hat{\theta}_{\text{MAP}} - \theta_{\text{sim}}$. Each tests an independent property: Bartlett information identity, Gaussian moment-matching, framework-vs.-empirical covariance agreement, and bias significance respectively.
- **Universal threshold.** The three matrix tests share a universal sampling-noise threshold $\tau_{\mathcal{D}} = 2\sqrt{p/(B-1)} \approx 0.49$ for $p = 6$ parameters and $B = 100$ bootstrap replicates. The Wald test uses the $\chi_{p,0.95}^2 = 12.59$ threshold. Tests exceeding their thresholds are flags; their pattern identifies the failure mode.
- **Two independent estimators of Σ .** The matrix-based DCC $\Sigma_{\text{DCC}}^{\text{post}}$ (computed via $\mathbf{H}_{\text{post}}$ and $\mathbf{J}\mathbf{T}_{\text{post}}$ at the global MAP) and the empirical bootstrap covariance $\hat{\Sigma}_{\text{emp}}$ (computed via per-replicate MAP optimisation) are asymptotically equivalent under correct algorithm specification. The $\mathbf{C}_{\text{match}}$ diagnostic tests their agreement in matrix form via the canonical $\mathcal{D}(\mathbf{C}_{\text{match}})$ statistic.
- **Composability.** The compositional gate (14) combines the four tests multiplicatively; any single threshold violation flags the algorithm. In production use, where θ_{sim} is unknown, T^2 is unavailable and the gate reduces to the three matrix tests.
- **$\mathbf{C}_{\text{overfit}}$ omitted** with the Cox-Snell magnitude argument (16): $\mathcal{D}(\mathbf{C}_{\text{overfit}}) \sim 10^{-6}$ at our effective sample size, seven orders of magnitude below the noise threshold.
- **Empirical case study.** macro_R at $\Delta t = 1\tau$ exhibits a categorical departure from macro_IR on the same data: $\lambda_{\min}(\mathbf{H}_{\text{lik}}) \approx -42$ in 100/100 replicates, coincident with $\text{rank}(\mathbf{G}_{\text{lik}}) < p$ in the same direction. macro_IR shows no such pattern at any Δt .

- **The fourth regime.** The empirical signature (18) — $\mathbf{G}_{\text{lik}}$ rank-deficient AND $\mathbf{H}_{\text{lik}}$ negative-eigenvalued coincident in the same direction — is a regime of algorithm failure distinct from classical misspecification, off-MLE bias, and weakly-identified directions. It is detectable directly from the PID spectral diagnostics and signals that the algorithm’s moment-matching approximation has structurally failed for the parameter configuration under test.
- **Production applicability.** The validation framework applies to real experimental data without knowledge of θ_{sim} : the per-replicate MAP optimisation and the matching test are both computable from data alone.
- **Computational cost.** Per-replicate MAP optimisation adds approximately a $2\times$ factor over global MAP alone, within the available cluster budget for figure_2 and tractable for experimental analyses of comparable scale.